

Thus, we have

$$H(X) \leq H_{\frac{1}{2}}(X) = \log(1 + Z(X)).$$

It follows that, for any jointly distributed pair (X, Y) with X binary and any sample value $Y = y$

$$H(X|Y = y) \leq \log(1 + Z(X|Y = y)).$$

Averaging over Y and by Jensen's inequality, we obtain (5).

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